



# Activities & Assessments

## Programming and Artificial Intelligence

CHAPTER 1 | PART ONE | FIRST SEMESTER

### Unit 1: Information Technology and Society

4 lessons | Performance tasks | Home assignments | Weekly assessments

## CHAPTER 1

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## Unit 1: Information Technology and Society

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### HOW THIS WORKBOOK IS ORGANISED

1

**Classroom Performances** (*continuous formative assessment*)

Varied activities the teacher uses during the lesson to boost student interaction, check progress directly, and identify strengths and areas needing improvement.

2

**Home Assignments** (*mastery and self-application*)

Guided activities and exercises that help students learn independently, check their understanding, and apply what they have learned in new contexts.

3

**Weekly Assessments** (*performance measurement and follow-up*)

Assessment models carried out at the end of each week to track progress and measure the learning outcomes achieved during that week.

**Part I** of each period contains the classroom *Performance Tasks*; **Part II** contains the *Home Assignments*. Period 3 of every lesson holds the *Weekly Assessments* in three models (A, B and C). Each question has dotted lines for your written answer.

## UNIT 1 | INFORMATION TECHNOLOGY AND SOCIETY

## LESSON

## 1

# The Evolution of Information Technology and Social Transformation

## PERIOD 1

Week: .....

Date: ...../...../.....

**Part I | Performance Tasks****Essay Questions:**

1. Explain the development of information technology from the 1940s to the 1960s, clarifying its primary use during that period.  
.....  
.....
2. Explain what is meant by “Moore’s Law” and whether it is considered a fixed physical law. Explain your answer.  
.....  
.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which of the following represents the main technology or event associated with the 1970s and 1980s?  

|                                  |                                                                               |
|----------------------------------|-------------------------------------------------------------------------------|
| A. The emergence of smartphones  | B. The spread of personal computers (PCs)                                     |
| C. The spread of cloud computing | D. The opening of the internet to commercial use and the emergence of the web |
2. Which empirical observation describes a historical trend in which the number of transistors on integrated circuits roughly doubles every two years?  

|                    |                          |
|--------------------|--------------------------|
| A. Cloud computing | B. Moore’s Law           |
| C. Edge computing  | D. Quantum superposition |

**Part II | Home Assignments****Essay Questions:**

1. State three social changes resulting from information technology.  
.....  
.....
2. What is meant by the term “e-commerce”? Illustrate your answer with two examples.  
.....  
.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which term refers to a work style in which a person performs their tasks from home or another remote location using the internet?
  - A. Online learning
  - B. E-commerce
  - C. Remote work
  - D. Cashless payment
2. What is meant by “social networking sites (SNS)”?
  - A. Platforms that allow users to communicate and to publish and share content quickly.
  - B. Buying and selling goods and services over the internet.
  - C. Paying for goods or services by non-cash means.
  - D. An educational model in which lessons are delivered over the internet.

**PERIOD 2**

Week: ..... Date: ...../...../.....

**Part I | Performance Tasks****Essay Questions:**

1. Explain how “edge computing” contributes to improving safety and decision-making speed in self-driving technology without the need to send data to the cloud.  
.....  
.....
2. Briefly compare “augmented reality (AR)” and “virtual reality (VR)” in terms of definition and how they work.  
.....  
.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which technology processes data on the device itself immediately, instead of sending it to the cloud, to avoid delays in decision-making?
  - A. Cloud computing
  - B. Quantum computing
  - C. Edge computing
  - D. Social networking sites
2. Which of the following represents the accurate concept of “augmented reality (AR)”?
  - A. A technology that places the user inside a fully computer-generated virtual environment.
  - B. A technology that adds digital elements or information to a scene from the real world.
  - C. A computing approach that uses the properties of quantum mechanics to process information.
  - D. A system for making payments using electronic money and QR codes.

**Part II | Home Assignments****Essay Questions:**

1. Explain the principle on which the “qubit” is based in quantum computing and how it differs from the “classical bit”.  
.....  
.....
2. How has the spread of cloud computing, from the second decade of the millennium onwards, changed the way IT resources are delivered?  
.....  
.....

**Multiple-Choice Questions:** (choose the correct answer)

1. What distinguishes the “qubit” from the “classical bit” in quantum computing?
  - A. It holds one specific state, either 0 or 1, at any given time.
  - B. It uses the principle of quantum superposition to be a mixture of 0 and 1 at the same time.
  - C. It relies solely on traditional integrated circuits and a stable electric current.
  - D. Its use is limited exclusively to simple calculations.
2. What is meant by the term “cashless payment”?
  - A. Selling and buying goods and services through physical stores using paper money.
  - B. Paying for goods or services by non-cash means such as cards, phone apps, or QR codes.
  - C. Working from home using the internet and email.
  - D. Delivering lessons and educational materials to users over the internet.

**PERIOD 3****Weekly Assessments**

Week: .....

Date: ...../...../.....

**MODEL A****Essay Questions:**

1. Explain the historical stages of the development of information technology, from the emergence of electronic computers in the 1940s to the 1960s, stating the primary purpose of their use during that period.  
.....  
.....
2. Explain “Moore’s Law”, clarifying what it describes and the engineering and physical challenges facing the continued miniaturization of circuit components.  
.....  
.....

3. List the five societal transformations resulting from information technology, and explain the concept of each of “remote work” and “e-commerce”.

4. Explain the characteristics and uses of the three prominent emerging technologies mentioned in the lesson (self-driving, augmented/virtual reality, and quantum computing).

**Multiple-Choice Questions:** (choose the correct answer)

1. In which period did the internet become available for commercial use and the web emerge, expanding global access to information?

- A. The 1940s and 1960s
- B. The 1990s
- C. The 1970s and 1980s
- D. From the second decade of the millennium onwards

2. Which technology uses AI to help drive a vehicle, using cameras and sensors to perceive the surroundings and make decisions?

- A. Cloud computing
- B. Self-driving
- C. Virtual reality
- D. Quantum computing

3. Which term describes information technology delivered as a service over the internet to support big-data analysis and AI?

- A.** Cloud computing
- B.** Edge computing
- C.** Classical bit
- D.** Quantum superposition

4. Which of the following is the correct description of the concept of “virtual reality (VR)”?

- A.** A technology that overlays digital information on images of the real world.
- B.** A technology that places the user inside a computer-generated virtual environment.
- C.** A technology for processing data on the device itself immediately, without the cloud.
- D.** A system for making payments using electronic money and QR codes.

## MODEL B

### Essay Questions:

1. Explain how the emergence of personal computers (PCs), global access to information, and email affected society in the 1970s and 1980s.

2. Explain how the continued miniaturization of circuit components under Moore's Law leads to engineering and physical challenges such as leakage currents, quantum effects, and quantum tunneling.

3. Briefly explain the concept of each of “social networking sites (SNS)”, “online learning”, and “cashless payment” as societal transformations.

.....

.....

4. Explain how edge computing processes data locally on board a self-driving vehicle, immediately, to reduce response time and ensure safety.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- The period that saw the emergence of smartphones and the spread of fast mobile internet is:  

|                        |                                       |
|------------------------|---------------------------------------|
| A. The 1960s and 1970s | B. The first decade of the millennium |
| C. The 1990s           | D. The 1940s                          |
- A technology that adds digital elements or information to a scene from the real world is known as:  

|                           |                      |
|---------------------------|----------------------|
| A. Virtual reality (VR)   | B. Self-driving      |
| C. Augmented reality (AR) | D. Quantum computing |
- Which of the following observations accurately describes what “Moore’s Law” does?  

|                                                                                        |                                                                          |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| A. The number of transistors on an integrated circuit roughly doubles every two years. | B. The number of transistors stays constant and never changes over time. |
| C. The energy consumption of conventional computers doubles every year.                | D. Computing and data-processing capabilities decline over the years.    |
- An educational and training model in which lessons and learning materials are delivered wholly or partly over the internet is:  

|                    |                     |
|--------------------|---------------------|
| A. E-commerce      | B. Remote work      |
| C. Online learning | D. Cashless payment |

**MODEL C**

**Essay Questions:**

1. Discuss the impact of the emergence of the web and the spread of the internet in the 1990s on making information available for commercial and global use.

.....

.....

2. Explain how self-driving is among the prominent emerging technologies.

.....

.....

3. Explain the concept of each of e-commerce and cashless payment.

.....

.....

4. Explain the fundamental differences between the “classical bit” and the “qubit”, using the concept of “quantum superposition” in quantum computing.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Electronic computers began to appear and were used mainly for military purposes and scientific calculations, such as the ENIAC computer, which used vacuum tubes, in the period of:  

|              |                                       |
|--------------|---------------------------------------|
| A. The 1940s | B. The 1970s                          |
| C. The 1990s | D. The first decade of the millennium |
2. A computing principle or approach that uses the properties of quantum mechanics to process information, and may offer an advantage in specific classes of problems, is:  

|                     |                            |
|---------------------|----------------------------|
| A. Edge computing   | B. Quantum computing       |
| C. Cashless payment | D. Social networking sites |
3. Platforms that allow users to communicate and to publish and share content quickly, and that are very effective at spreading information, are:  

|                                  |                    |
|----------------------------------|--------------------|
| A. Social networking sites (SNS) | B. E-commerce      |
| C. Remote work                   | D. Cloud computing |
4. A work style in which a person performs their tasks and duties from home or another remote location, using communication technologies and the internet, is called:  

|                    |                   |
|--------------------|-------------------|
| A. Online learning | B. Remote work    |
| C. Self-driving    | D. Edge computing |

## UNIT 1 | INFORMATION TECHNOLOGY AND SOCIETY

## LESSON

## 2

## How Artificial Intelligence Works

## PERIOD 1

Week: .....

Date: ...../...../.....

## Part I | Performance Tasks

## Essay Questions:

1. Define the concept of artificial intelligence, explaining in three points the difference between machine learning, deep learning, and generative AI.

.....

.....

2. Explain how models learn in the branch of “machine learning”, compared with traditional methods based on explicitly programming rules.

.....

.....

## Multiple-Choice Questions: (choose the correct answer)

1. Which of the following represents the broad field that includes computer systems able to perform tasks such as learning, prediction, and recognizing speech and images?

A. Deep learning

B. Artificial intelligence

C. Machine learning

D. Generative AI

2. What is “machine learning” from the perspective of the sequence of AI technologies?

A. A branch of AI in which models learn patterns from data.

B. An approach that relies exclusively on disconnected neural networks.

C. A system for generating images and video clips exclusively, without training data.

D. A set of rules that are manually programmed in a strict and fixed way.

## Part II | Home Assignments

## Essay Questions:

1. What are the basic components of an artificial neural network?

.....

.....

2. Explain the concept of “generative AI (GenAI)”, illustrating it with examples.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- Which term describes a machine-learning approach that relies on multi-layered neural networks to learn complex representations and patterns?
  - Machine learning
  - Generative AI
  - Deep learning
  - Spam filter
- What is the core function of “generative AI” systems?
  - Explicitly programming rules for old computers.
  - Creating new content such as text, images, audio, and software.
  - Classifying spam emails only, without generating content.
  - Operating integrated circuits according to Moore’s Law.

**PERIOD 2**

Week: ..... Date: ...../...../.....

**Part I | Performance Tasks****Essay Questions:**

- Explain the relationship between artificial intelligence, machine learning, deep learning, and generative AI.

.....

.....

- Explain what is meant by the term “hallucination” in the context of generative AI.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- Which of the following technologies do most modern generative AI systems rely on to create new content?
  - Explicitly programmed rules.
  - Deep learning models and neural networks.
  - Edge computing and Moore’s Law.
  - Fixed classical bits.
- What is meant by the term “hallucination” when generative AI produces text?
  - Producing text that seems plausible and logical but is factually incorrect.
  - Stopping responding completely because of an internet outage.
  - Generating images that are one hundred percent accurate and perfectly match reality.
  - Classifying spam emails with extreme accuracy.

**Part II | Home Assignments****Essay Questions:**

- Suppose a farmer wants to use an AI system to distinguish images of healthy crops from diseased ones. Explain the type of data the system needs for training, and identify one factor that could make its predictions wrong.

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.....

2. Discuss one benefit and one risk of using generative AI to write a school report, and what learners should do to verify the outputs.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which of the following is NOT a suitable example of “generative AI” applications?
- A. The ChatGPT model for generating text.                      B. AI image-generation tools.
- C. Spam filter.                                                              D. Audio and sound-clip generation tools.
2. What are the basic components of an artificial neural network that enable it to learn complex patterns from data?
- A. Connected layers containing neuron units whose weights change during training.                      B. A set of vacuum tubes and simple electric circuits.
- C. Fixed software rules written as purely linear algorithms.                                                              D. Manual switches to regulate the flow of electric current in the processor.

PERIOD 3

Weekly Assessments

Week: .....

Date: ...../...../.....

MODEL A

**Essay Questions:**

1. Define artificial intelligence and explain how today’s intelligent systems differ from traditional computers programmed with fixed rules.
- .....
- .....
2. Explain in detail the overlapping relationship between artificial intelligence, machine learning, and deep learning, supporting your answer with the basic concepts of each.
- .....
- .....
3. What is “generative AI”, and how does it rely on deep learning models to create new content such as text and images?
- .....
- .....
4. Discuss the problem of “hallucination” in generative AI, explaining why it is dangerous and the appropriate procedure for verifying outputs.
- .....
- .....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which of the following represents a branch of AI in which models learn patterns from data to make predictions or classifications without explicitly programming every rule?
  - A. Deep learning
  - B. Generative AI
  - C. Machine learning
  - D. Cloud computing
2. A machine-learning approach that relies on multi-layered neural networks to learn complex representations and patterns, such as image analysis and speech recognition, is:
  - A. Deep learning
  - B. Artificial intelligence
  - C. Generative AI
  - D. Traditional programming
3. A computer system loosely inspired by the idea of interconnected neurons, whose weights change during training, is:
  - A. The quantum qubit
  - B. The artificial neural network
  - C. The classical bit
  - D. Moore's Law
4. When generative AI produces text that seems plausible but is factually incorrect, this phenomenon is called:
  - A. Hallucination
  - B. Quantum superposition
  - C. Edge computing
  - D. Linear operation

**MODEL B****Essay Questions:**

1. Explain how everyday examples such as spam classification, product recommendations, and translation help in understanding AI applications.

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.....
2. Clearly compare "machine learning" and "deep learning" in terms of reliance on neural networks and the amount of data required for training.

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.....
3. Explain how generative AI can create new and varied content such as text, images, and sounds based on training data.

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.....
4. Set an appropriate rule for handling generative AI in school assignments, giving two convincing reasons why the accuracy of sources must be verified.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which of the following is a direct example of “machine learning” applications?
  - A. Spam filters and product recommendations
  - B. Generating complex artistic images and videos
  - C. Processing data on the edge devices of cars
  - D. Operating integrated circuits according to Moore’s Law
2. What is the accurate description of “generative AI (GenAI)”?
  - A. A system limited to classifying spam mail and removing viruses.
  - B. AI that uses deep learning to generate new data such as text and images.
  - C. A computing technology that uses quantum mechanics and parallel processing.
  - D. A simple neural network with only one input layer and one output layer.
3. What are the layers located between the input layer and the output layer in an artificial neural network called?
  - A. Hidden layers
  - B. Edge layers
  - C. Cloud layers
  - D. Linear layers
4. What should learners do to avoid the risks of “hallucination” when using generative AI to prepare their school reports?
  - A. Rely absolutely on the outputs without any human review.
  - B. Verify the outputs and their sources and compare them with reliable facts.
  - C. Avoid using phones and computers completely and permanently.
  - D. Copy the texts word for word and hand them straight to the teacher.

**MODEL C****Essay Questions:**

1. Why are most current AI systems considered “narrow” (Narrow AI)?

.....

.....
2. Define the artificial neural network.

.....

.....
3. Draw a simple diagram showing the relationship between artificial intelligence, deep learning, machine learning, and generative AI.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Most current AI systems are designed to perform:
  - A. A specific task or a limited set of tasks (narrow AI).
  - B. Simulating full human consciousness in all fields without limits.
  - C. Manual programming of old household electrical appliances.
  - D. Managing global military communication networks only.

2. Which of the following is the most suitable technology for complex image-analysis tasks and advanced speech recognition?
- A. Deep learning based on multi-layered neural networks.
  - B. Manually programmed rules in a fixed linear style.
  - C. Traditional cloud computing without AI.
  - D. E-commerce and cashless payment.
3. What is the main benefit of the speed that generative AI provides to students when preparing reports?
- A. Helping to complete tasks quickly, while paying attention to the risk of inaccuracy.
  - B. Automatically guaranteeing that all information is one hundred percent correct.
  - C. Completely doing away with official school books and sources.
  - D. Completely preventing the phenomenon of hallucination.
4. What is the proper procedure to follow when using generative AI tools in assignments and projects?
- A. Always verifying the accuracy of information and reviewing the original sources and references.
  - B. Treating everything the screens produce as absolute scientific facts that cannot be questioned.
  - C. Not writing any school research and relying entirely on AI.
  - D. Submitting AI outputs without reading or examining them.

## UNIT 1 | INFORMATION TECHNOLOGY AND SOCIETY

## LESSON

## 3

## Artificial Intelligence in Daily Life and Industry

## PERIOD 1

Week: ..... Date: ...../...../.....

## Part I | Performance Tasks

## Essay Questions:

1. Explain how technologies such as video platforms and voice assistants have become part of everyday services, and what the essential role of AI is in them.

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.....

2. Explain what is meant by “AI for image-based diagnosis” in healthcare, and how it contributes to detecting diseases.

.....

.....

## Multiple-Choice Questions: (choose the correct answer)

1. Which of the following is an example of a system that predicts preferences from past behavior data and displays recommendations?  

|                          |                     |
|--------------------------|---------------------|
| A. Recommendation system | B. Voice assistant  |
| C. Machine translation   | D. Face recognition |
2. Which service automatically translates text into different languages, as mentioned in everyday AI services?  

|                          |                     |
|--------------------------|---------------------|
| A. Recommendation system | B. Voice assistant  |
| C. Machine translation   | D. Face recognition |

## Part II | Home Assignments

## Essay Questions:

1. List four essential AI services in daily life, stating the role of AI in each service.

.....

.....

2. What role does AI play in the manufacturing sector? Give two illustrative examples.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- Which term refers to the AI system that recognizes voice, understands commands, and carries them out, such as Siri and Google Assistant?
  - Recommendation system
  - Voice assistant
  - Machine translation
  - Face recognition
- Which of the following industrial sectors is associated with the task of “optimizing delivery routes”?
  - Healthcare
  - Agriculture
  - Manufacturing
  - Logistics

**PERIOD 2**

Week: ..... Date: ...../...../.....

**Part I | Performance Tasks****Essay Questions:**

- Explain how AI is used in the agriculture sector and its role in monitoring crops and pests.

.....

.....

- State four tasks that AI can perform in processing complex data and recognizing images and sounds.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- Which industry uses AI to detect diseases from X-ray and computed tomography (CT) images?
  - Healthcare
  - Agriculture
  - Manufacturing
  - Logistics
- Which of the following represents the main task of AI in the agriculture sector?
  - Optimizing delivery routes
  - Predicting harvest time and detecting pests and diseases
  - Product quality inspection and predictive maintenance
  - Analyzing medical images and discovering drugs

**Part II | Home Assignments****Essay Questions:**

- Explain the saying: “AI is good at finding patterns in data, but human judgment is still necessary.”

.....

.....

2. State three precautions for use related to AI.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which term describes the generation of incorrect or unsupported information that seems convincing by AI?
  - A. The black box problem
  - B. Hallucination
  - C. Discrimination or bias
  - D. Insufficient training data
2. What is meant by the “black box” problem in the context of AI applications?
  - A. Generating incorrect information that seems convincing to the user.
  - B. A lack of clarity about how the AI reached its judgment or decision.
  - C. Using works protected by copyright as training data.
  - D. A shortage of training data, or its failure to represent reality accurately.

PERIOD 3

Weekly Assessments

Week: .....

Date: ...../...../.....

MODEL A

**Essay Questions:**

1. Explain how recommendation systems are used in daily life, with applied examples such as YouTube and Amazon.

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2. State the uses of AI in healthcare.

.....

.....
3. State four major industrial sectors into which AI technologies are being introduced, briefly describing the role AI plays in each.

.....

.....
4. Explain what AI excels at and what requires caution when using it, with regard to decisions with an ethical dimension and to privacy.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- Which of the following services is used to unlock a smartphone and to detect and automatically recognize people's faces in photos?
  - Recommendation system
  - Voice assistant
  - Machine translation
  - Face recognition
- Which technology is used for product quality inspection and for predicting machine failures before they occur?
  - AI for image-based diagnosis
  - AI in manufacturing and predictive maintenance
  - Recommendation systems and voice assistants
  - Machine translation and social networks
- Among the precautions for using AI, what does the phenomenon of "hallucination" mean?
  - Not knowing how the system reached its decision.
  - Generating incorrect or unsupported information that seems convincing.
  - The presence of bias in the training data.
  - Violating the copyright of data.
- Which of the following is the correct description of what AI excels at?
  - Finding and classifying patterns in complex data, and probabilistic inference.
  - Making absolute ethical judgments without humans.
  - Guaranteeing that outputs are completely free of any future bias.
  - Storing personal data and protecting it with complete confidentiality.

**MODEL B****Essay Questions:**

- State the role of AI in voice assistants, giving two examples.

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.....
- Discuss the role of AI in the agriculture and logistics sectors in improving operations and productivity.

.....

.....
- Briefly explain the precautions for use associated with biased training data, high-impact decisions, and determining responsibility.

.....

.....
- Explain why confirmation from a human doctor is necessary for medical diagnosis, instead of relying on AI alone.

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

- A daily service used to understand speech and carry out voice commands, such as Siri and Google Assistant, is:
  - Recommendation system
  - Voice assistant
  - Machine translation
  - Face recognition

2. An industrial sector that uses AI to optimize delivery routes is:
  - A. Healthcare
  - B. Agriculture
  - C. Manufacturing
  - D. Logistics
3. Which term describes the lack of clarity about how AI reached its judgment or decision?
  - A. Hallucination
  - B. The black box problem
  - C. Discrimination or bias
  - D. Probabilistic inference
4. Among the tasks that AI performs well:
  - A. Making complex ethical decisions.
  - B. Recognizing images, sounds, and texts, and generating content.
  - C. Guaranteeing absolute accuracy of results without training data.
  - D. Protecting personal privacy automatically.

### MODEL C

#### Essay Questions:

1. Explain three services that AI can perform in daily life.  
.....  
.....
2. State two examples of AI's contribution to the manufacturing sector.  
.....  
.....
3. Explain the dimensions related to the black box problem and to rights issues when works protected by copyright are used as training data.  
.....  
.....
4. Explain the fundamental differences between what AI excels at (finding patterns) and the aspects that require great caution, such as privacy and ethics.  
.....  
.....

#### Multiple-Choice Questions: (choose the correct answer)

1. A system that predicts preferences from past behavior data and displays recommendations on platforms such as YouTube and Amazon is called:
  - A. Recommendation system
  - B. Voice assistant
  - C. Machine translation
  - D. Face recognition
2. Which of the following industries uses AI to predict harvest time and to detect pests and diseases?
  - A. Healthcare
  - B. Agriculture
  - C. Manufacturing
  - D. Logistics

3. Decisions with an ethical dimension that may lead to discrimination or bias are classified under:
- A. What AI excels at
  - B. What requires caution when using AI
  - C. The core skills of deep learning
  - D. The role of everyday recommendation systems
4. Bias in AI systems is defined as:
- A. A pattern that may lead to unfair results
  - B. Hallucination
  - C. Quantum superposition
  - D. Probabilistic inference

## UNIT 1 | INFORMATION TECHNOLOGY AND SOCIETY

## LESSON

## 4

## Ethical Issues Related to Artificial Intelligence

## PERIOD 1

Week: .....

Date: ...../...../.....

## Part I | Performance Tasks

## Essay Questions:

1. What is meant by the term “algorithmic bias” in AI systems, and what are its most prominent forms or examples?

.....

.....

2. Explain the main causes of algorithmic bias in AI systems.

.....

.....

## Multiple-Choice Questions: (choose the correct answer)

1. Which of the following is an inappropriate or incorrect cause of algorithmic bias?

A. Biased training data

B. The reflection of past discriminatory tendencies in the data

C. Slow computer processing speed

D. Insufficient data about certain characteristics

2. What does the term “algorithmic bias” represent in AI systems?

A. A pattern in AI outputs that may lead to unfair or harmful results.

B. A technology that lets humans understand why AI reached a particular judgment.

C. Appropriate handling and protection of personal data.

D. Identifying the parties responsible for the system, its decisions, and its effects.

## Part II | Home Assignments

## Essay Questions:

1. Explain how the development of AI technology has led to new issues related to privacy and surveillance in public places.

.....

.....

2. What is meant by the term “training data” in AI systems, and how does insufficient representation affect the outputs?

.....

.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which of the following represents the accurate concept of “training data”?
- A. The examples from which an AI system learns its patterns before it is used.
  - B. A technology that lets humans understand why AI reached a particular judgment.
  - C. Identifying the parties responsible for the system, its decisions, and its effects.
  - D. Not discriminating unjustly against any person or group.
2. What is one of the privacy issues associated with AI?
- A. Improvement in the computing speed of AI
  - B. AI creating creative works
  - C. Identifying and tracking individuals through facial recognition technology
  - D. Increased efficiency of microprocessors

PERIOD 2

Week: .....

Date: ...../...../.....

**Part I | Performance Tasks**

**Essay Questions:**

1. What is meant by “explainable AI (XAI)”, and why is it considered a fundamental approach to overcoming the black box problem?
- .....
- .....
2. Briefly compare the concept of “responsibility” with the concept of “accountability” in the context of AI systems.
- .....
- .....

**Multiple-Choice Questions:** (choose the correct answer)

1. Which term describes the technology that lets humans understand the factors and the reason that led AI to reach a particular judgment?
- A. Cloud computing
  - B. Explainable AI (XAI)
  - C. Edge computing
  - D. Algorithmic bias
2. What is the basic difference between “responsibility” and “accountability”, as explained in the lesson?
- A. Responsibility means showing the decision-making process, while accountability means processing speed.
  - B. Responsibility relates to defining the roles and duties of the parties, while accountability means identifying the parties that can be held to account according to their roles.
  - C. Accountability is the technology that understands the logic of AI, while responsibility is data collection.
  - D. There is no difference between them; they are completely synonymous terms.

**Part II | Home Assignments****Essay Questions:**

1. State the four basic principles of AI ethics, with a brief explanation of the principle of “fairness”.  
.....  
.....
2. Explain what is meant by each of the principle of “transparency” and the principle of “privacy protection” within the framework of responsible AI use.  
.....  
.....

**Multiple-Choice Questions:** (choose the correct answer)

1. Among the basic principles of AI ethics, which principle requires not discriminating unjustly against any person or specific group?  
A. Transparency  
B. Accountability  
C. Fairness  
D. Privacy protection
2. Which principle refers to making clear and appropriate information available about the system, its use, its decision-making process, and its limits?  
A. Fairness  
B. Transparency  
C. Privacy protection  
D. Responsibility

**PERIOD 3****Weekly Assessments**

Week: .....

Date: ...../...../.....

**MODEL A****Essay Questions:**

1. Explain what “algorithmic bias” in AI systems is, showing how it arises from training data or from the design of the system and the way it is used, with examples.  
.....  
.....
2. Discuss in detail “explainable AI (XAI)” and the “black box” problem, and how it contributes to evaluating the system and discovering errors.  
.....  
.....
3. Compare precisely the concept of “responsibility” with the concept of “accountability”, explaining why the distribution of responsibility differs across contexts and parties (developer, operator, user).  
.....  
.....

4. State the four basic principles of AI ethics and briefly explain what is meant by each of “fairness”, “transparency”, “privacy protection”, and “accountability”.

**Multiple-Choice Questions:** (choose the correct answer)

1. A pattern in AI outputs that may lead to unfair or harmful results, and that arises from data or design, is called:
  - A. Explainable AI (XAI)
  - B. Algorithmic bias
  - C. Absolute transparency
  - D. Privacy protection
2. The technology that lets humans understand why AI reached a particular judgment, and that helps avoid the black box problem, is:
  - A. Explainable AI (XAI)
  - B. Edge computing
  - C. Training data
  - D. Algorithmic bias
3. The principle that means handling personal information appropriately and protecting it is:
  - A. Fairness
  - B. Transparency
  - C. Privacy protection
  - D. Accountability
4. Identifying the parties responsible for the system, its decisions, and its effects, and being able to hold them to account according to their roles, represents the principle of:
  - A. Transparency
  - B. Fairness
  - C. Accountability
  - D. Privacy protection

## MODEL B

### Essay Questions:

1. Explain how unrepresentative or biased training data (such as the reflection of past discriminatory tendencies) can lead AI to repeat this bias in its judgments.

2. Explain the “black box” problem in complex AI algorithms, and why it makes evaluating the mechanism and discovering errors difficult without the use of explainability techniques.

3. State the privacy issues associated with the spread of facial recognition technologies in public places and with the mass collection of data and behavior across the internet.

4. Explain in detail the four basic principles of AI ethics and how each of them guides the responsible use of these technologies in society.

**Multiple-Choice Questions:** (choose the correct answer)

1. The examples from which an AI system learns its patterns before it is used, and which directly affect the quality of its outputs, are called:
  - A. Training data
  - B. Encryption algorithms
  - C. Edge devices
  - D. Random-access memory
2. Making clear and appropriate information available about the system, its use, its decision-making process, and its limits is an accurate definition of the principle of:
  - A. Fairness
  - B. Transparency
  - C. Accountability
  - D. Privacy protection
3. Not discriminating unjustly against any person or specific group in the judgments and outputs of AI represents the principle of:
  - A. Fairness
  - B. Transparency
  - C. Privacy protection
  - D. Accountability
4. The method or technology that helps humans understand the factors that contributed to the system reaching a particular output or decision is:
  - A. Algorithmic bias
  - B. Explainable AI (XAI)
  - C. Mass data collection
  - D. Network auditing

## MODEL C

### Essay Questions:

1. Discuss the causes of algorithmic bias and how it can arise from the inappropriate selection of variables, or from using a proxy variable that indirectly represents a protected attribute.
2. Show the importance of “explainable AI (XAI)” in addressing the difficulty of evaluating how a result was reached when it is not clear to the human eye.
3. Explain the different dimensions of the responsibility of multiple parties (developer, operating company, user) when an AI system makes a wrong or unfair judgment.

4. Explain how the four basic principles of AI ethics (fairness, transparency, privacy protection, accountability) work together to form a complete framework for responsible use.
- .....
- .....

**Multiple-Choice Questions:** (choose the correct answer)

1. Defining the roles and duties of the parties involved in developing, operating, and using the system (developer, operator, user) refers to the term:
  - A. Responsibility
  - B. Algorithmic bias
  - C. Training data
  - D. Transparency
2. A system or technology that collects large amounts of behavior data across the internet and allows individuals to be tracked in public places raises issues related to:
  - A. Computer processing speed
  - B. Privacy protection
  - C. Central memory efficiency
  - D. Moore's Law
3. Which of the following represents the accurate description of the principle of "accountability"?
  - A. Making clear information about the system and its limits available.
  - B. Not discriminating unjustly against any person or group.
  - C. Identifying the parties responsible for the system and its decisions, and being able to hold them to account according to their roles.
  - D. Understanding the factors that contributed to the system reaching a particular output.
4. Which of the following is a clear example of bias in AI systems?
  - A. A hiring system that unfairly favors one group, or a facial recognition system whose accuracy is lower for certain groups.
  - B. A program for organizing personal folders and files on a local computer.
  - C. An application for calculating simple mathematical and geometric operations.
  - D. A tool for editing landscape photos and improving their lighting.